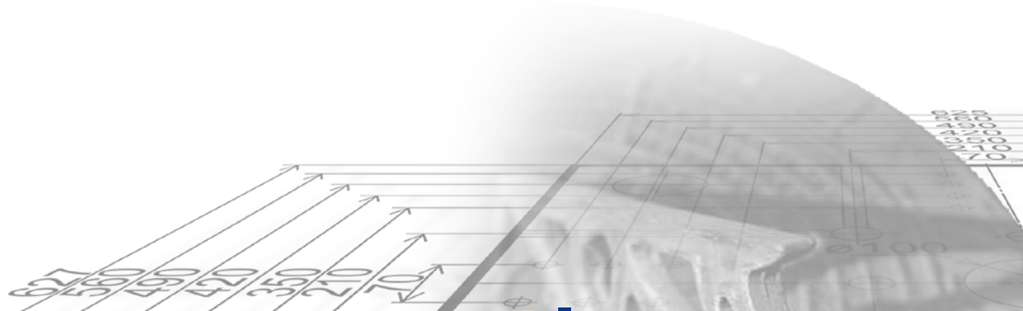


Tianjian **Pluto** Series

TIANJIAN
PLASTICS MACHINERY



# Tianjian **Pluto** Series
## Specifications
860-10000KN




**HAITIAN INTERNATIONAL HOLDINGS LIMITED**
Unit 1105 Level 11
Metroplaza Tower 2
223 Hing Fong RD Kwai Fong N.T
haitian@mail.haitian.com
www.haitian.com

**NINGBO HAITIAN Tianjian MACHINERY CO., LTD.**
Export Processing Zone, Ningbo 315800 China
+86-574-86 177 242
+86-574-86 221 864
haitian@mail.haitian.com
www.haitian.com


**Tianjian Partner:**

HT 20150605-IV

A Member of Haitian International

We Create Advantage.



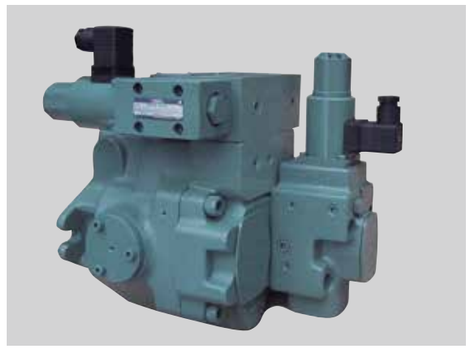
# Tianjian Pluto Series

## *is an economical solution for energy saving*

The Pluto series machine was designed to provide an economical solution for energy saving. Utilizing a proportional variable pump hydraulic drive system, the machine performance is highly responsive giving a repeatable process window for the part production. A power saving of 30 ~ 50% is possible , when compared to traditional hydraulic drive systems. Because the drive system uses "on-demand" oil flow technology, the oil cooling is significantly reduced lowering the machine water consumption, which extends the life of the machine hydraulic components. Haitian is committed to providing intelligent energy saving solution without performance compromise to the injection molding industry.



The hydraulic drive system provides an economical energy saving solution, without sacrificing machine performance.

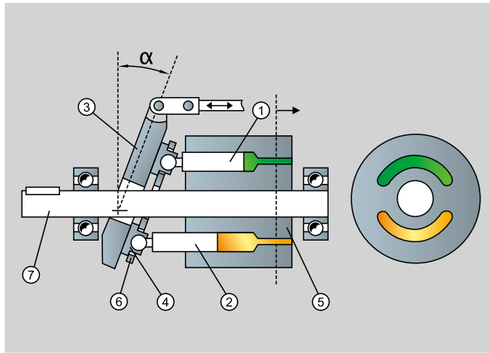


Proportional variable displacement piston pump

### Main Features:

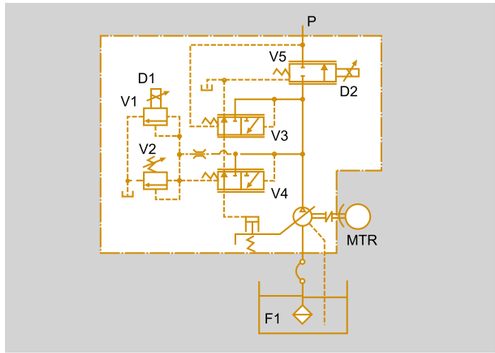
- Economic energy saving hydraulic drive system
- Optimized design for rapid response and repeatable control
- Intelligent user interface for convenient mold setting
- Quality components for extended machine life
- Excellent safety system for machine operation and operator protection
- Quality Screw and Barrels for processing different materials.

### Variable Pump Design



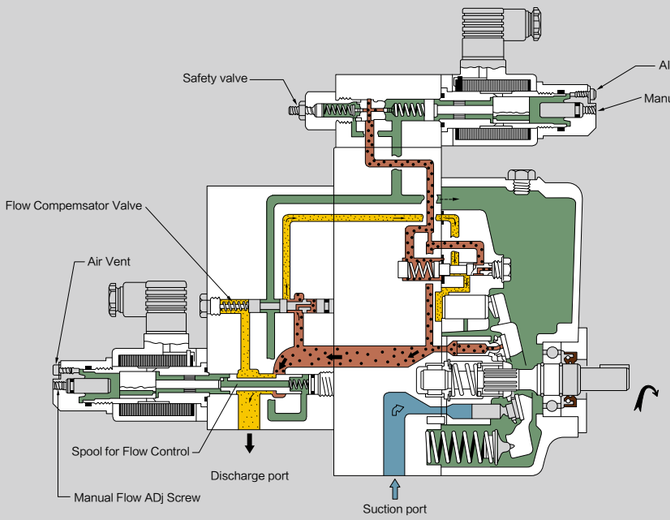

The constant speed motor rotates the drive shaft (7) to turn the angular rotating plate (3). Connected to the plate are pistons (1)(2) which change their oil flow according the (α) angle of the rotating plate. Energy saving is achieved by changing the oil flow required for each machine movement.

### Drive System Principle




V1: proportional pressure valve
V2: safety valve
V3: flow compemsator valve
V4: pressure compemsator valve
V5: proportional flow valve

### Diagram of proportional variable pump valve

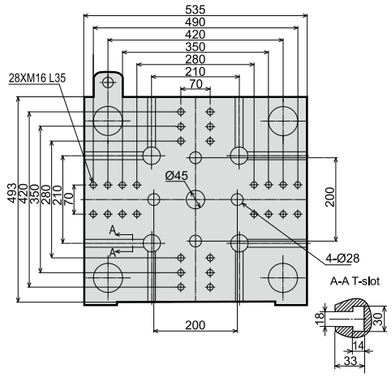


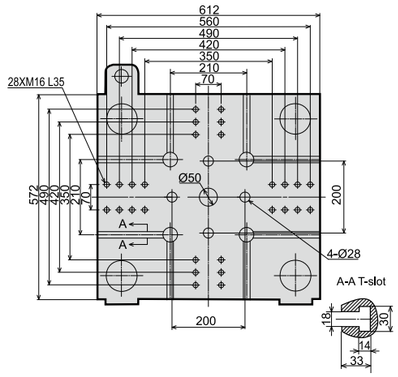
# Specification

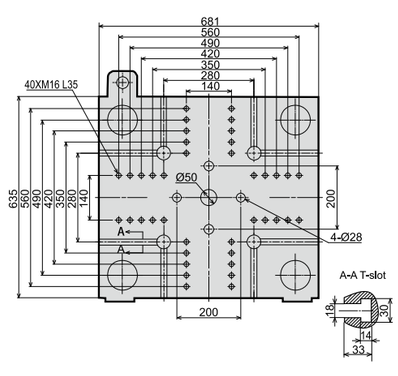
| | | PL860/260 | | | PL1200/370 | | | PL1600/540 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **INJECTION UNIT** | | A | B | C | A | B | C | A | B | C |
| Screw Diameter | mm | 32 | 36 | 40 | 36 | 40 | 45 | 40 | 45 | 50 |
| Screw L /D Ratio | L/D | 22.5 | 20 | 18 | 23.3 | 21 | 18.7 | 22.5 | 20 | 18 |
| Shot Size (Theoretical) | $cm^3$ | 121 | 153 | 188 | 173 | 214 | 270 | 253 | 320 | 395 |
| Injection Weight (PS) | g | 110 | 139 | 172 | 157 | 194 | 246 | 230 | 291 | 359 |
| Injection Rate | g/s | 66 | 84 | 104 | 83 | 102 | 130 | 100 | 126 | 156 |
| Injection Pressure | Mpa | 237 | 187 | 151 | 231 | 187 | 148 | 239 | 189 | 153 |
| Plasticizing Capacity(ps) | g/s | 9.9 | 12.4 | 14.6 | 12.4 | 14.6 | 19.5 | 14.6 | 19.5 | 24 |
| Screw Speed | rpm | | 0-205 | | | 0-180 | | | 0-175 | |
| **CLAMPING UNIT** | | | | | | | | | | |
| Clamp Tonnage | KN | | 860 | | | 1200 | | | 1600 | |
| Toggle Stroke | mm | | 300 | | | 340 | | | 400 | |
| Space Between Tie Bars | mm | | 360×315 | | | 405×365 | | | 455×410 | |
| Max.Mold Height | mm | | 305 | | | 380 | | | 435 | |
| Min. Mold Height | mm | | 125 | | | 130 | | | 155 | |
| Ejector Stroke | mm | | 75 | | | 100 | | | 100 | |
| Ejector Tonnage | KN | | 30 | | | 40 | | | 40 | |
| **OTHERS** | | | | | | | | | | |
| Max. Pump Pressure | Mpa | | 17.5 | | | 17.5 | | | 17.5 | |
| Pump Motor Power | Kw | | 7.5 | | | 11 | | | 15 | |
| Heater Power | Kw | | 6.2 | | | 9.75 | | | 9.75 | |
| Machine Dimension(L×W×H) | m | | 4.0×1.08×1.9 | | | 4.39×1.15×1.98 | | | 4.62×1.21×2.02 | |
| Machine Weight | t | | 3.1 | | | 4.2 | | | 4.8 | |
| Hopper Capacity | Kg | | 25 | | | 25 | | | 25 | |
| Oil Tank Capacity | L | | 165 | | | 190 | | | 245 | |

**Platen dimensions**

Moving platen

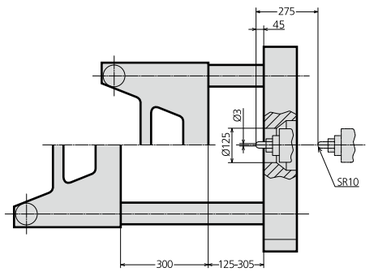


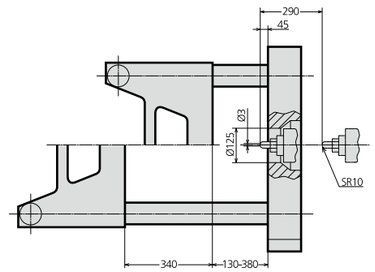



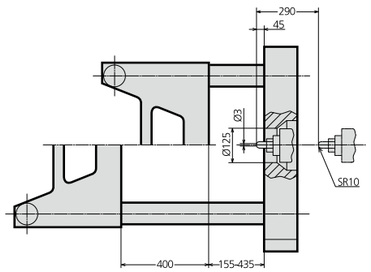




**Platen dimensions**

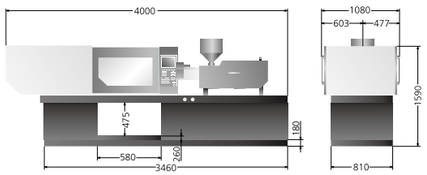
Mounting hole for robot/sprue picker top view from fixed platen

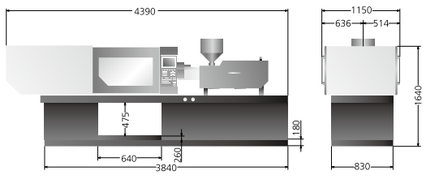


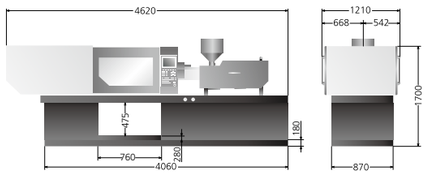




**Machine dimensions**



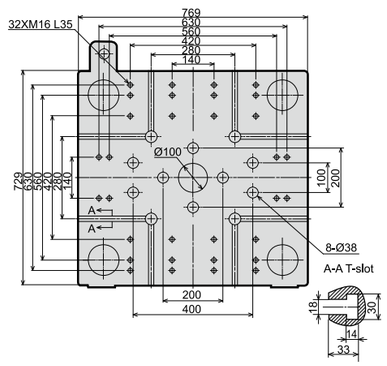

We reserve the right to make changes as a result of further technical advantages.

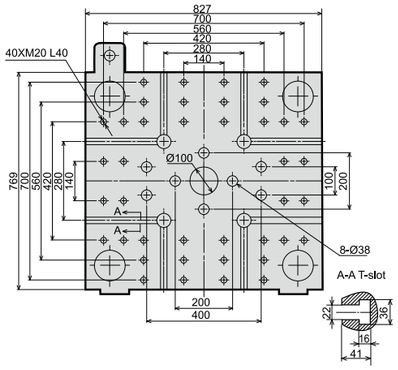
# Specification

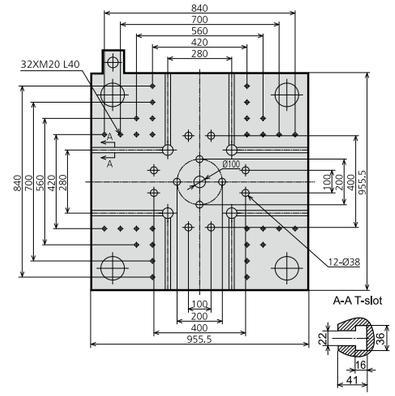
| | | PL2000/770 | | | PL2500/1000 | | | PL3000/1800 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **INJECTION UNIT** | | A | B | C | A | B | C | A | B | C |
| Screw Diameter | mm | 45 | 50 | 55 | 50 | 55 | 60 | 60 | 65 | 70 |
| Screw L /D Ratio | L/D | 22.2 | 20 | 18.2 | 22 | 20 | 18.3 | 21.7 | 20 | 18.6 |
| Shot Size (Theoretical) | $cm^3$ | 366 | 452 | 546 | 497 | 601 | 715 | 845 | 992 | 1151 |
| Injection Weight (PS) | g | 333 | 411 | 497 | 452 | 547 | 651 | 769 | 903 | 1047 |
| Injection Rate | g/s | 158 | 195 | 235 | 167 | 202 | 241 | 234 | 274 | 318 |
| Injection Pressure | Mpa | 213 | 172 | 142 | 204 | 169 | 142 | 213 | 181 | 156 |
| Plasticizing Capacity(ps) | g/s | 19.5 | 24 | 29.6 | 24 | 29.6 | 33.7 | 33.7 | 38.2 | 44.9 |
| Screw Speed | rpm | | 0-165 | | | 0-175 | | | 0-185 | |
| **CLAMPING UNIT** | | | | | | | | | | |
| Clamp Tonnage | KN | | 2000 | | | 2500 | | | 3000 | |
| Toggle Stroke | mm | | 450 | | | 510 | | | 600 | |
| Space Between Tie Bars | mm | | 505×465 | | | 560×500 | | | 660×660 | |
| Max.Mold Height | mm | | 480 | | | 530 | | | 660 | |
| Min. Mold Height | mm | | 160 | | | 180 | | | 250 | |
| Ejector Stroke | mm | | 140 | | | 140 | | | 160 | |
| Ejector Tonnage | KN | | 62 | | | 62 | | | 62 | |
| **OTHERS** | | | | | | | | | | |
| Max. Pump Pressure | Mpa | | 17.5 | | | 16 | | | 16 | |
| Pump Motor Power | Kw | | 18.5 | | | 22 | | | 30 | |
| Heater Power | Kw | | 14.25 | | | 16.65 | | | 19.65 | |
| Machine Dimension(L×W×H) | m | | 5.2×1.31×2.1 | | | 5.69×1.39×2.15 | | | 6.4×1.76×2.45 | |
| Machine Weight | t | | 6.1 | | | 7.4 | | | 11.5 | |
| Hopper Capacity | Kg | | 50 | | | 50 | | | 50 | |
| Oil Tank Capacity | L | | 305 | | | 395 | | | 550 | |

**Platen dimensions**

Moving platen

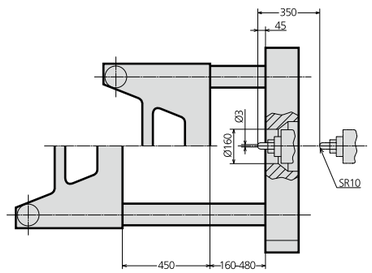


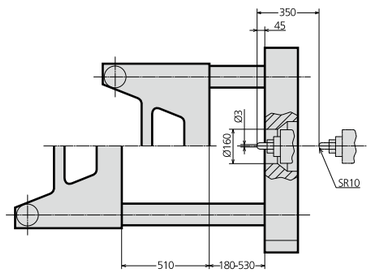



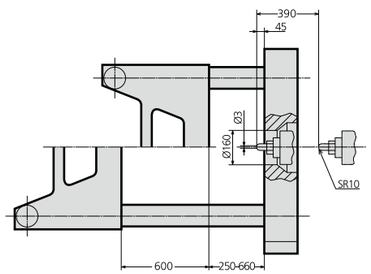




**Platen dimensions**

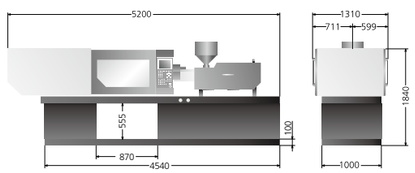
Mounting hole for robot/sprue picker top view from fixed platen

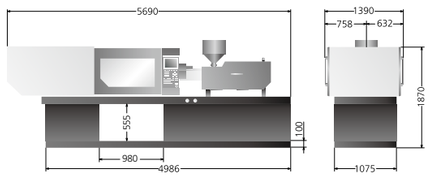


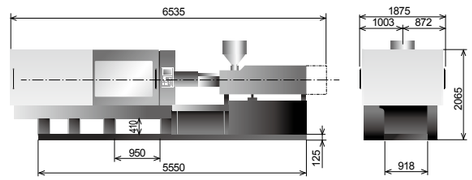




**Machine dimensions**



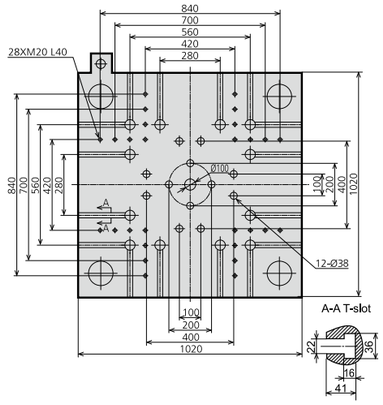

We reserve the right to make changes as a result of further technical advantages.

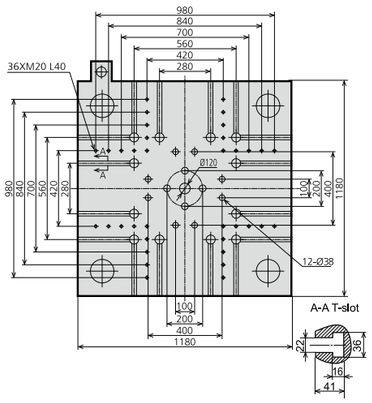
# Specification

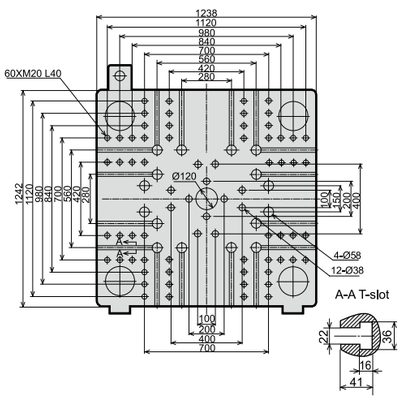
| | | PL3600/2250 | | | PL4500/2900 | | | PL5300/4000 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **INJECTION UNIT** | | A | B | C | A | B | C | A | B | C | D |
| Screw Diameter | mm | 65 | 70 | 80 | 70 | 80 | 90 | 80 | 85 | 90 | 100 |
| Screw L /D Ratio | L/D | 21.5 | 20 | 17.5 | 22.9 | 20 | 17.8 | 22 | 20.7 | 19.6 | 17.6 |
| Shot Size (Theoretical) | $cm^3$ | 1068 | 1239 | 1619 | 1424 | 1860 | 2354 | 2212 | 2497 | 2799 | 3456 |
| Injection Weight (PS) | g | 972 | 1128 | 1473 | 1296 | 1692 | 2142 | 2013 | 2272 | 2547 | 3145 |
| Injection Rate | g/s | 318 | 369 | 481 | 351 | 458 | 580 | 450 | 508 | 569 | 703 |
| Injection Pressure | Mpa | 211 | 182 | 139 | 203 | 156 | 123 | 199 | 176 | 157 | 127 |
| Plasticizing Capacity(ps) | g/s | 38.2 | 44.9 | 58.7 | 44.9 | 58.7 | 71.8 | 58.7 | 65.4 | 71.8 | 86.5 |
| Screw Speed | rpm | | 0-185 | | | 0-160 | | | 0-145 | | |
| **CLAMPING UNIT** | | | | | | | | | | | |
| Clamp Tonnage | KN | | 3600 | | | 4500 | | | 5300 | | |
| Toggle Stroke | mm | | 660 | | | 740 | | | 825 | | |
| Space Between Tie Bars | mm | | 710×710 | | | 780×780 | | | 830×800 | | |
| Max.Mold Height | mm | | 710 | | | 780 | | | 850 | | |
| Min. Mold Height | mm | | 250 | | | 330 | | | 350 | | |
| Ejector Stroke | mm | | 160 | | | 200 | | | 200 | | |
| Ejector Tonnage | KN | | 110 | | | 110 | | | 158 | | |
| **OTHERS** | | | | | | | | | | | |
| Max. Pump Pressure | Mpa | | 17.5 | | | 16 | | | 17.5 | | |
| Pump Motor Power | Kw | | 37 | | | 45 | | | 55 | | |
| Heater Power | Kw | | 24.85 | | | 29.45 | | | 40.25 | | |
| Machine Dimension(L×W×H) | m | | 7.4×1.88×2.6 | | | 7.95x2.05x2.7 | | | 9.45×2.3×2.62 | | |
| Machine Weight | t | | 14.6 | | | 18.5 | | | 25 | | |
| Hopper Capacity | Kg | | 50 | | | 100 | | | 200 | | |
| Oil Tank Capacity | L | | 640 | | | 820 | | | 990 | | |

**Platen dimensions**

Moving platen

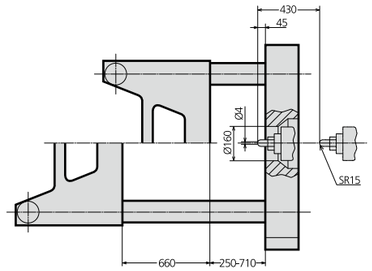


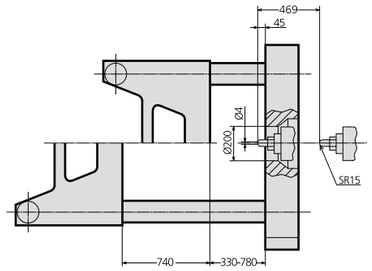



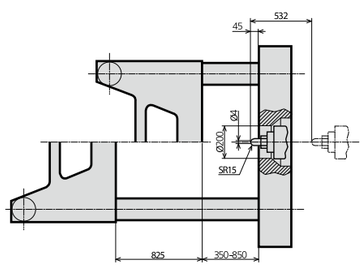




**Platen dimensions**

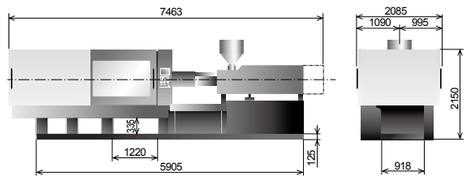
Mounting hole for robot/sprue picker top view from fixed platen

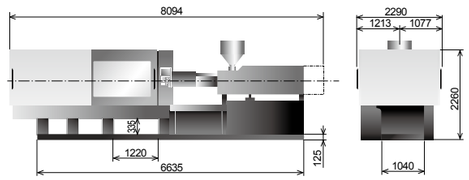


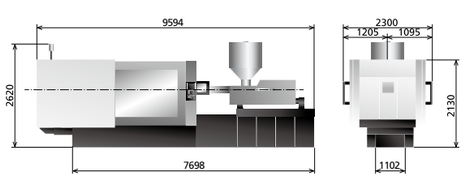




**Machine dimensions**



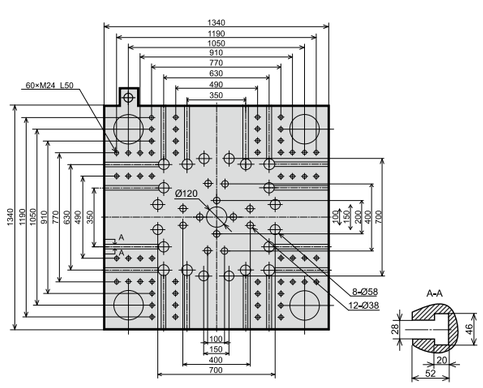

We reserve the right to make changes as a result of further technical advantages.

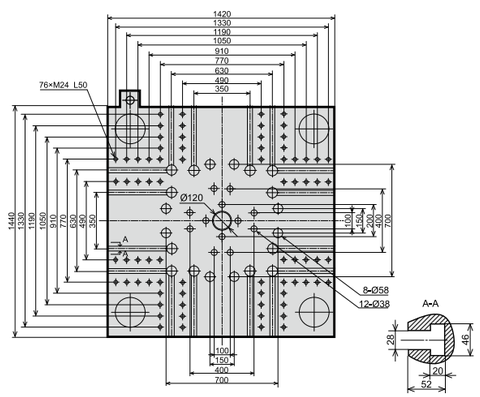
# Specification

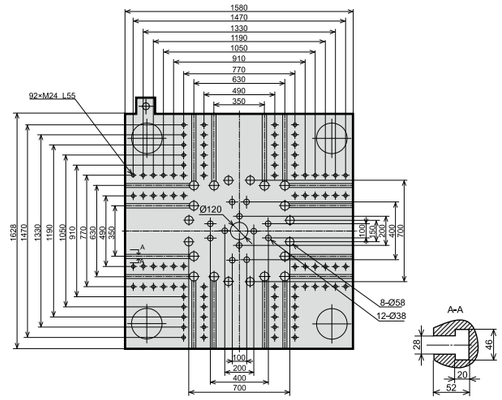
| | | PL6500/4550 | | | | PL7800/6350 | | | | PL10000/7950 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **INJECTION UNIT** | | A | B | C | D | A | B | C | D | A | B | C | D |
| Screw Diameter | mm | 80 | 90 | 100 | 110 | 90 | 100 | 110 | 120 | 100 | 110 | 120 | 130 |
| Screw L /D Ratio | L/D | 24.8 | 22 | 19.8 | 18 | 24.4 | 22 | 20 | 18.3 | 24.2 | 22 | 20.2 | 18.6 |
| Shot Size (Theoretical) | $cm^3$ | 2036 | 2576 | 3181 | 3849 | 2799 | 3456 | 4181 | 4976 | 3770 | 4562 | 5429 | 6371 |
| Injection Weight (PS) | g | 1853 | 2345 | 2895 | 3502 | 2547 | 3145 | 3805 | 4528 | 3431 | 4151 | 4940 | 5798 |
| Injection Rate | g/s | 417 | 528 | 651 | 788 | 531 | 655 | 793 | 944 | 760 | 920 | 1095 | 1285 |
| Injection Pressure | Mpa | 223 | 177 | 143 | 118 | 234 | 190 | 157 | 132 | 220 | 182 | 153 | 130 |
| Plasticizing Capacity(ps) | g/s | 58.7 | 71.8 | 86.5 | 104.0 | 71.8 | 86.5 | 104.0 | 122.0 | 86.5 | 104.0 | 122.0 | 139.0 |
| Screw Speed | rpm | 0-120 | | | | 0-119 | | | | 0-116 | | | |
| **CLAMPING UNIT** | | | | | | | | | | | | | |
| Clamp Tonnage | KN | 6500 | | | | 7800 | | | | 10000 | | | |
| Toggle Stroke | mm | 900 | | | | 980 | | | | 1100 | | | |
| Space Between Tie Bars | mm | 895×895 | | | | 980×980 | | | | 1090×1090 | | | |
| Max.Mold Height | mm | 900 | | | | 980 | | | | 1100 | | | |
| Min. Mold Height | mm | 400 | | | | 400 | | | | 500 | | | |
| Ejector Stroke | mm | 260 | | | | 260 | | | | 320 | | | |
| Ejector Tonnage | KN | 175 | | | | 186 | | | | 215 | | | |
| **OTHERS** | | | | | | | | | | | | | |
| Max. Pump Pressure | Mpa | 16 | | | | 17.5 | | | | 17.5 | | | |
| Pump Motor Power | Kw | 22+37 | | | | 37+37 | | | | 37+55 | | | |
| Heater Power | Kw | 49.25 | | | | 61.15 | | | | 68.75 | | | |
| Machine Dimension(L×W×H) | m | 9.9×2.32×2.7 | | | | 10.5×2.42×2.73 | | | | 11.4×2.68×2.81 | | | |
| Machine Weight | t | 30.2 | | | | 36 | | | | 47.5 | | | |
| Hopper Capacity | Kg | 200 | | | | 200 | | | | 200 | | | |
| Oil Tank Capacity | L | 1300 | | | | 1360 | | | | 1610 | | | |

**Platen dimensions**

Moving platen

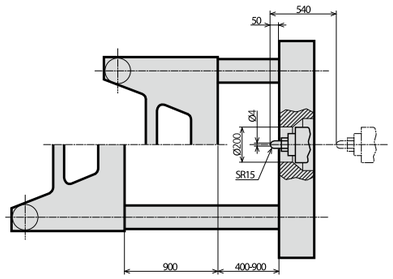


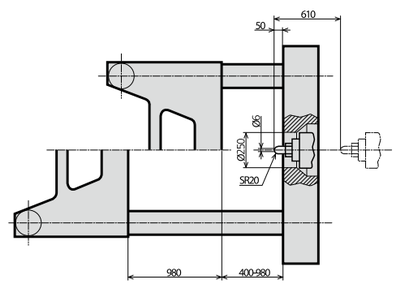



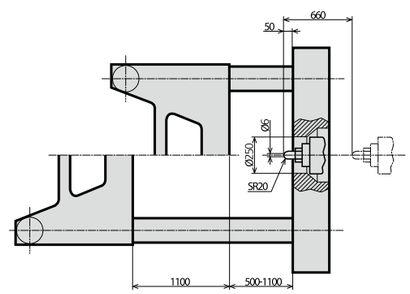




**Platen dimensions**

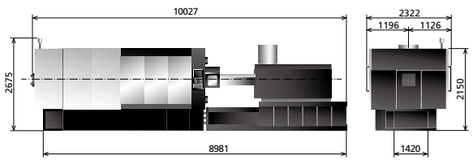
Mounting hole for robot/sprue picker top view from fixed platen

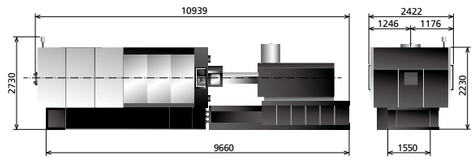


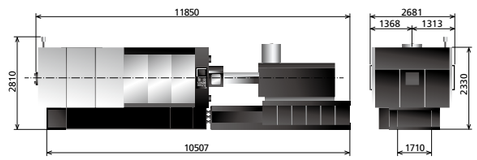




**Machine dimensions**




We reserve the right to make changes as a result of further technical advantages.